

LAPORAN PRATIKUM BASIS DATA

Penerapan Subquery dan Operator SQL pada tabel Employees



Disusun Oleh:

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KATA PENGANTAR

Puji dan syukur kita panjatkan kehadirat Allah yang Maha Esa karena telah memberikan nikmat sehat jasmani dan rohani sehingga saya bisa menyelesaikan laporan ini. Di dalam laporan ini, terdapat hasil praktikum, pembahasan serta kesimpulan yang saya susun secara sistematis. Saya sebagai penulis sekaligus penyusun laporan ini sangat berterimakasih kepada Ibu Rahmi Eka Putri, MT selaku dosen dalam mata kuliah praktikum Basis Data departemen Informatika, Universitas Andalas, yang telah menuntun dan mengajarkan saya mata kuliah ini.

Dalam penulisan laporan praktikum ini, saya mendapatkan suatu pelajaran baik serta mendapatkan pengetahuan tentang praktikum tersebut. Saya sebagai penulis dalam laporan ini masih dalam proses belajar, baik dalam tata penulisan laporan yang benar maupun dalam memahami penyajian laporan yang sesuai kaidah. Untuk itu saya berharap mendapatkan saran-saran yang membangun sehingga dapat meningkatkan pengetahuan saya dan dapat membuat isi laporan yang lebih baik, mendidik dan dapat dipertanggungjawabkan.

Padang, 2026

Shofwatul Hasanah Sarma

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BAB I

PENDAHULUAN

1.1 Latar Belakang

Praktikum ini dilakukan untuk memahami penggunaan subquery dalam SQL pada pengolahan data basis data. Dengan subquery, pengguna mengambil data berdasarkan kondisi tertentu secara lebih spesifik dan efisien pada tabel database

1.2 Tujuan Pratikum

Tujuan dari pelaksanaan praktikum antara lain sebagai berikut :

1. Memahami konsep subquery dalam SQL
2. Melatih penggunaan query tunggal dan bertingkat
3. Menampilkan dan mengolah data berdasarkan kondisi tertentu pada database

1.3 Manfaat praktikum

manfaat dari pelaksanaan praktikum antara lain sebagai berikut :

1. Menambah pemahaman tentang pengolahan data menggunakan sql
2. Meningkatkan kemampuan membuat query database
3. Membantu mahasiswa memahami hubungan antar data dalam tabel

BAB II

PEMBAHASAN

2.1 Pengertian

Praktikum basis data tentang subquery merupakan kegiatan pembelajaran untuk mempelajari teknik query SQL yang menggunakan perintah SELECT di dalam SELECT lain guna memperoleh data yang lebih spesifik dan terstruktur

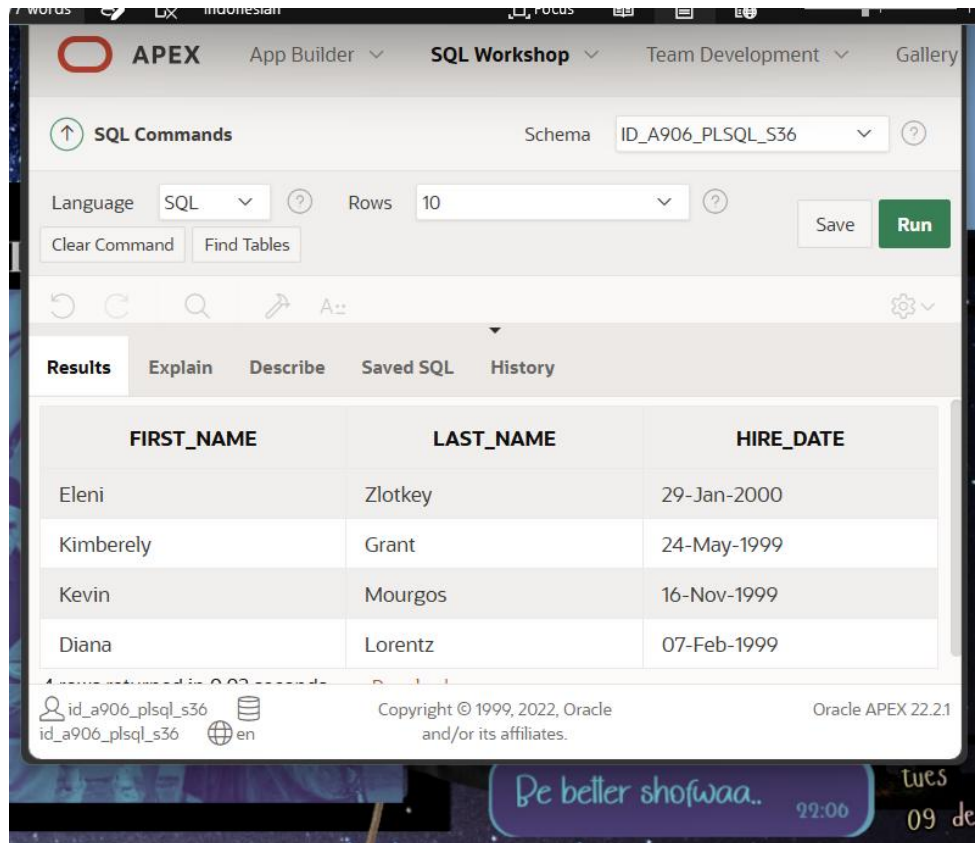
2.2 Pratikum

1. Section 10.1

a. Halaman 10

- Query:

```
select first_name, last_name, hire_date  
from employees  
where hire_date >  
(select hire_date  
from employees  
where last_name = 'Vargas');
```
- Penjelasan:
Menampilkan first name, last name dan hire date dari tabel employee dengan hire date ber last name = 'Vargas'
- Output:



b. Halaman 11

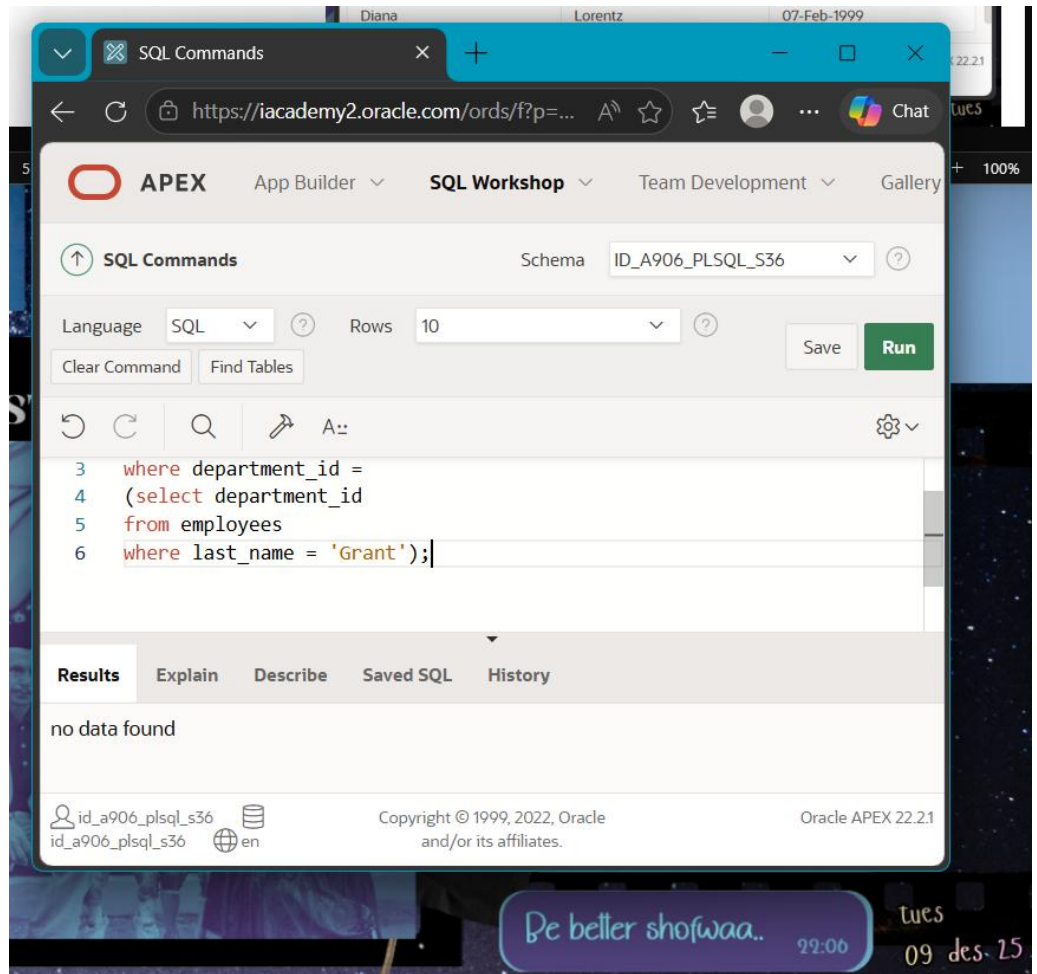
i. Query:

```
select last_name
from employees
where department_id =
(select department_id
from employees
where last_name = 'Grant');
```

ii. Penjelasan:

Menampilkan out time datang kepada mahasiswa. Terutama kepada sekali dbirrdilu..lojudash make upp..?

iii. Output:



2. Section 10.2

a. Halaman 7

i. query:

```
select last_name, job_id, department_id
from employees
where department_id = (select department_id
from departments
where department_name = 'Marketing')
order by job_id;
```

ii. penjelasan:

menampilkan last name, job id, department id dari tabel employee dimana department name = marketing

iii. output:

The screenshot shows the Oracle APEX SQL Workshop interface. The browser address bar displays the URL `https://iacademy2.oracle.com/ords/f?p=...`. The APEX header includes the APEX logo, navigation links for App Builder, SQL Workshop, Team Development, and Gallery. The main section is titled "SQL Commands" and shows the schema "ID_A906_PLSQL_S36". The language is set to "SQL" and the number of rows to display is "10". There are buttons for "Clear Command", "Find Tables", "Save", and "Run". Below the command area, there are tabs for "Results", "Explain", "Describe", "Saved SQL", and "History". The "Results" tab is active, showing a table with the following data:

LAST_NAME	JOB_ID	DEPARTMENT_ID
Hartstein	MK_MAN	20
Fay	MK_REP	20

Below the table, it states "2 rows returned in 0.02 seconds" and provides a "Download" link. The footer includes the user "id_a906_plsql_s36", the language "en", the copyright "Copyright © 1999, 2022, Oracle and/or its affiliates.", and the version "Oracle APEX 22.2.1".

b. Halaman 8

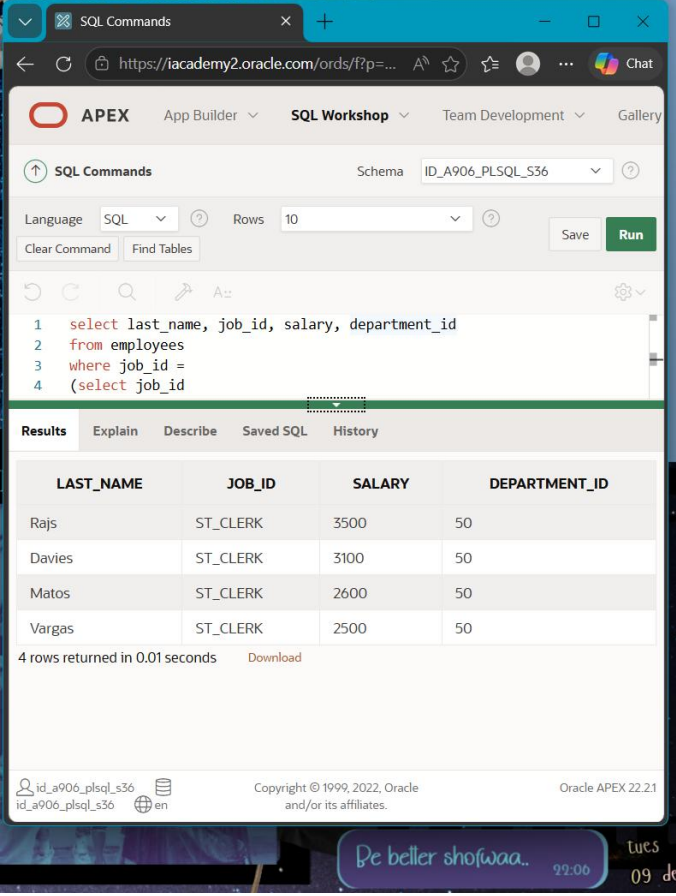
i. Query:

```
select last_name, job_id, salary, department_id
from employees
where job_id =
(select job_id
from employees
where employee_id = 141)
and department_id =
(select department_id
from departments
where location_id = 1500);
```

ii. Penjelasan:

Menampilkan last name, job id, salary dan department id dari tabel employee dimana job id dengan employee id = 141 dan department id dengan location id = 1500

iii. Output:



The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command entered is:

```
1 select last_name, job_id, salary, department_id
2 from employees
3 where job_id =
4 (select job_id
```

The results tab shows the following data:

LAST_NAME	JOB_ID	SALARY	DEPARTMENT_ID
Rajs	ST_CLERK	3500	50
Davies	ST_CLERK	3100	50
Matos	ST_CLERK	2600	50
Vargas	ST_CLERK	2500	50

4 rows returned in 0.01 seconds

c. Halaman 10

i. Query:

```
select last_name, salary
from employees
where salary <
(select AVG (salary)
from employees);
```

ii. Penjelasan:

Menampilkan last name dan salary dari tabel employee. Salary merupakan jumlah total dari last name.

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' section is active, showing a schema of 'ID_A906_PLSQL_S36'. The language is set to 'SQL' and the number of rows to display is '10'. The SQL command entered is:

```
1 select last_name, salary
2 from employees
3 where salary <
4 (select AVG (salary))
```

The results are displayed in a table with two columns: 'LAST_NAME' and 'SALARY'.

LAST_NAME	SALARY
Whalen	4400
Gietz	8300
Taylor	8600
Grant	7000
Mourgos	5800
Rajs	3500
Davies	3100
Mateo	2600

The footer of the interface shows the user 'id_a906_plsql_s36', the copyright notice 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'.

d. Halaman 12

i. Query:

```
select department_id, min (salary)
from employees
group by department_id
having min (salary)>
(select min (salary)
from employees
where department_id = 50);
```

ii. Penjelasan:

Menampilkan department id dengan salary minimum dari tabel employee.

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The main area is titled 'SQL Commands' and shows the schema 'ID_A906_PLSQL_S36'. The language is set to 'SQL' and the number of rows to display is '10'. The query entered is:

```
(select min (salary)
from employees
where department_id = 50);
```

The results are displayed in a table with two columns: 'DEPARTMENT_ID' and 'MIN(SALARY)'. The table contains 7 rows of data. Below the table, it states '7 rows returned in 0.01 seconds' and provides a 'Download' link. The footer includes the user 'id_a906_plsql_s36', the copyright 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'.

DEPARTMENT_ID	MIN(SALARY)
110	8300
90	17000
10	4400
-	7000
20	6000
60	4200
80	8600

3. Section 10.3

a. Halaman 8

i. Query:

```
select last_name, hire_date
from employees
where extract (year from hire_date) in
(select extract (year from hire_date)
from employees
where department_id=90);
```

ii. Penjelasan:

Menampilkan last name dengan hire date dari tabel employee. Di mana hire name berasal dari department id = 90

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command entered is:

```
(select extract (year from hire_date)
from employees
where department_id=90);
```

The results are displayed in a table with two columns: LAST_NAME and HIRE_DATE. The table contains four rows of data:

LAST_NAME	HIRE_DATE
King	17-Jun-1987
Whalen	17-Sep-1987
Kochhar	21-Sep-1989
De Haan	13-Jan-1993

Below the table, it states "4 rows returned in 0.01 seconds" and provides a "Download" link. The footer of the interface includes the user "id_a906_plsql_s36", the Oracle logo, and the text "Copyright © 1999, 2022, Oracle and/or its affiliates. Oracle APEX 22.2.1".

b. Halaman 10

i. Query:

```
select last_name, hire_date
from employees
where extract (year from hire_date)<any
(select extract (year from hire_date)
from employees
where department_id = 90);
```

ii. Penjelasan:

Menampilkan last name dengan hire date. Di mana hire datenya sebelum 90, tepat 90 dan sesudah 90

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The query entered is:

```
(select extract (year from hire_date)
from employees
where department_id = 90);
```

The results are displayed in a table with two columns: LAST_NAME and HIRE_DATE. The table contains 5 rows of data.

LAST_NAME	HIRE_DATE
Whalen	17-Sep-1987
King	17-Jun-1987
Kochhar	21-Sep-1989
Hunold	03-Jan-1990
Ernst	21-May-1991

5 rows returned in 0.01 seconds

c. Halaman 11

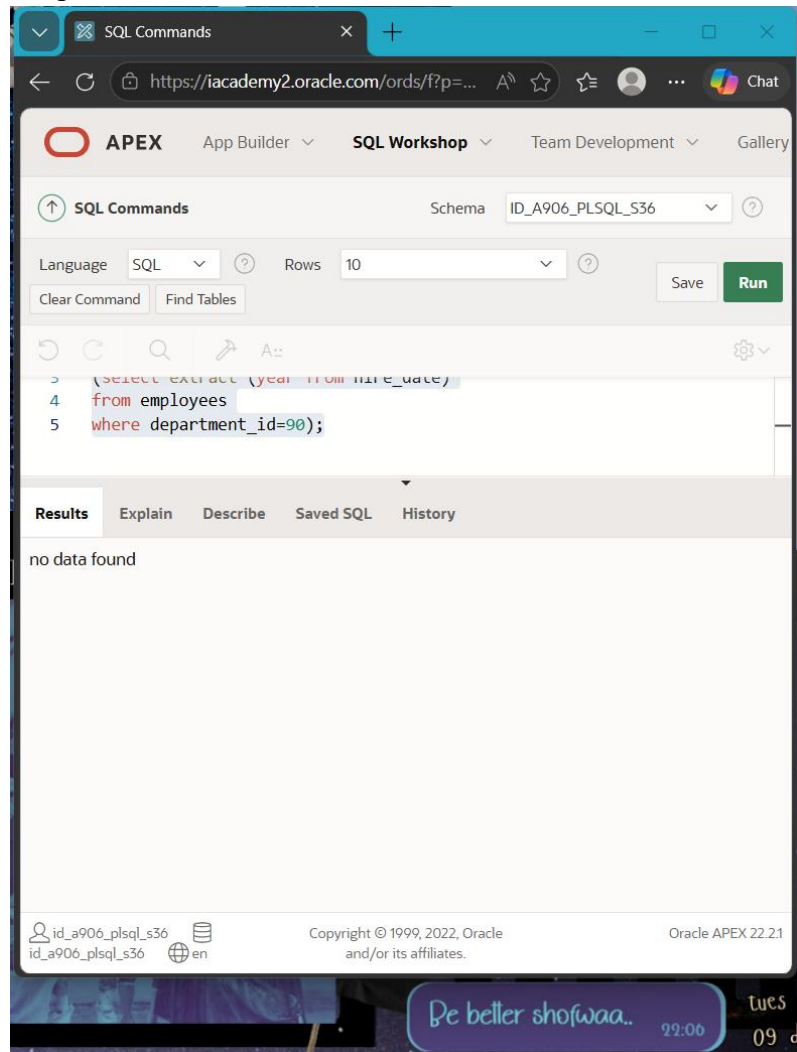
i. Query:

```
select last_name, hire_date from employees
where extract (year from hire_date) < all
(select extract (year from hire_date)
from employees
where department_id = 90);
```

ii. Penjelasan:

Menampilkan last name dari hire date dimana hire date berdepartment id = 90

iii. Output:



d. Halaman 12

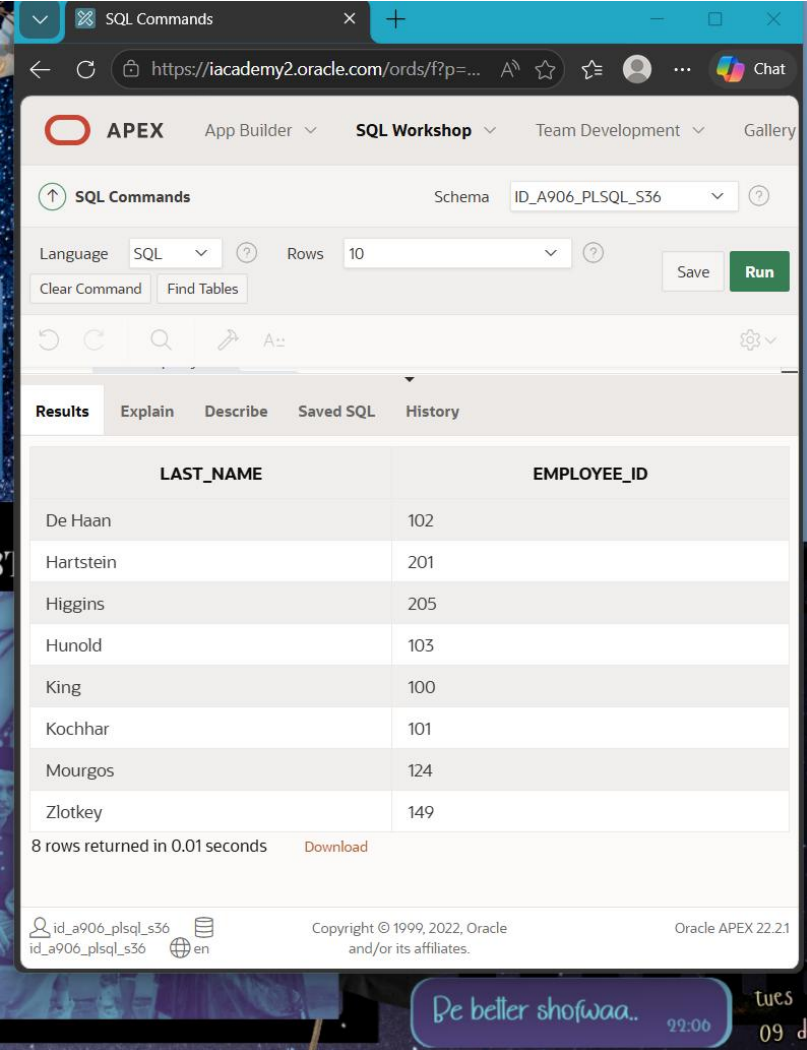
i. Query:

```
select last_name,
employee_id
from employees
where employee_id in
(select manager_id
from employees);
```

ii. Penjelasan:

Menampilkan last name dengan employee id, dengan manager id ada dalam tabel employee

iii. Output:



The screenshot shows the Oracle APEX SQL Workshop interface. The 'SQL Commands' tab is active, displaying a query result table. The table has two columns: 'LAST_NAME' and 'EMPLOYEE_ID'. The results show 8 rows of employee data. Below the table, it indicates '8 rows returned in 0.01 seconds' and provides a 'Download' link. The footer of the interface shows the user 'id_a906_plsql_s36', the copyright 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'.

LAST_NAME	EMPLOYEE_ID
De Haan	102
Hartstein	201
Higgins	205
Hunold	103
King	100
Kochhar	101
Mourgos	124
Zlotkey	149

e. Halaman 13

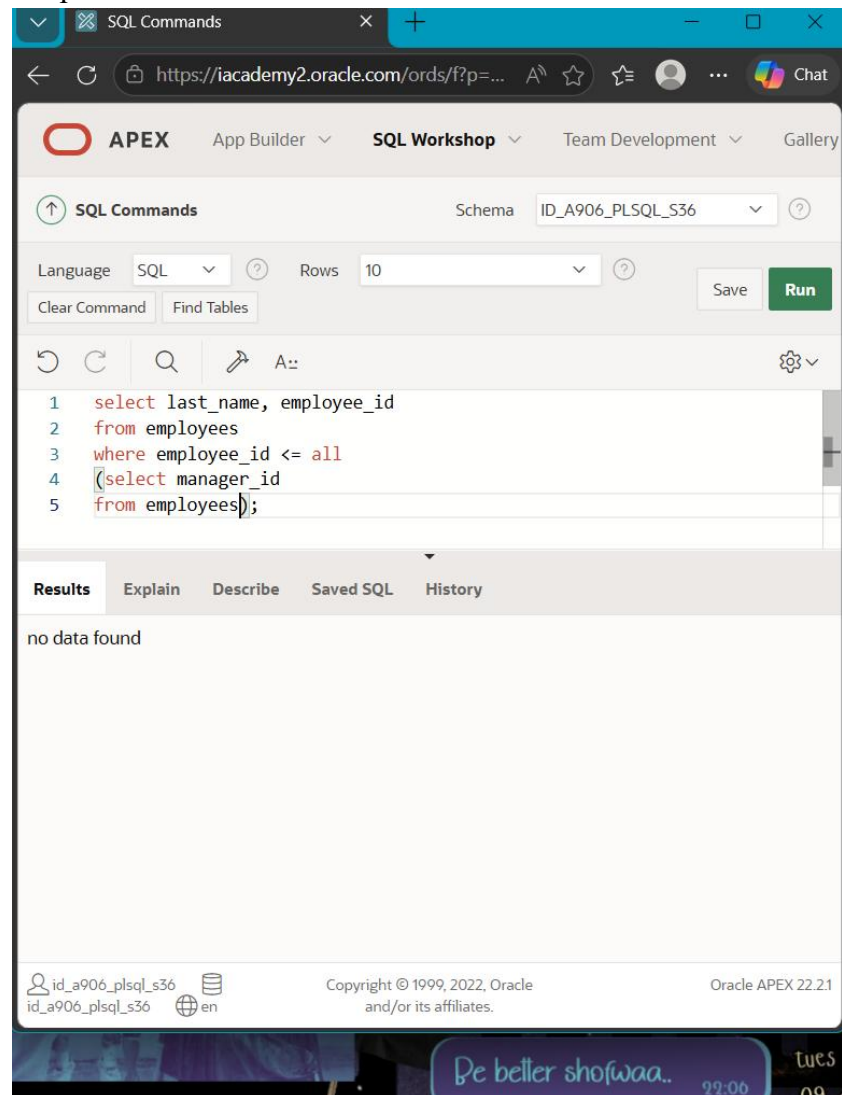
i. Query:

```
select last_name, employee_id
from employees
where employee_id <= all
(select manager_id
from employees);
```

ii. Penjelasan:

Menampilkan last name dan employee_id dengan employee id berasal dari semua manager id dari tabel employee

iii. Output:



f. Halaman 16

i. Query:

```
select department_id, MIN(salary)
from employees
group by department_id
having min(salary) < ANY
(select salary
from employees
where department_id IN (10,20))
```

order by department_id;

ii. Penjelasan:

menampilkan department id dengan min salary, di mana min salary berada di semua department id = (10,20)

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command entered is:

```
4 having min(salary) < ANY
5 (select salary
6 from employees
7 where department_id IN (10,20))
8 order by department_id;
```

The results are displayed in a table with two columns: DEPARTMENT_ID and MIN(SALARY).

DEPARTMENT_ID	MIN(SALARY)
10	4400
20	6000
50	2500
60	4200
80	8600
110	8300

The interface also shows the schema ID_A906_PLSQL_S36, the language SQL, and the number of rows set to 10. The footer indicates the copyright is © 1999, 2022, Oracle and/or its affiliates, and the version is Oracle APEX 22.2.1.

g. Halaman 17

i. Query:

```
select employee_id, manager_id, department_id
from employees
where (manager_id, department_id) in
(select manager_id, department_id
from employees
where employee_id in (149,174))
and employee_id not in (149, 174)
```

ii. Penjelasan:

Menampilkan employee id, manager id dan department_id dari tabel employee. Di mana manager id dan department id berasal dari tabel employee id in (149, 174)

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command entered is:

```
1 select employee_id, manager_id, department_id
2 from employees
3 where (manager_id, department_id) in
4 (select manager_id, department_id
5 from employees
6 where employee_id in (149,174))
```

The results tab shows one row of data:

EMPLOYEE_ID	MANAGER_ID	DEPARTMENT_ID
176	149	80

1 rows returned in 0.01 seconds

h. Halaman 19

i. Query:

```
select employee_id, manager_id, department_id
from employees
where manager_id in
(select manager_id
from employees
where employee_id in (149,174))
and department_id in
```

```
(select department_id
from employees
where employee_id in
(149, 174))
and employee_id not in (149, 174);
```

ii. Penjelasan:

Menampilkan employee id, manager id dan department id dari tabel employee. Dengan manager id berasal dari tabel employee id in (149, 174) dan department id berasal dari employee id in (149, 174)

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command entered is:

```
(select department_id
from employees
where employee_id in
(149, 174))
and employee_id not in (149, 174);
```

The results tab is active, displaying a table with the following data:

EMPLOYEE_ID	MANAGER_ID	DEPARTMENT_ID
176	149	80

Below the table, it states "1 rows returned in 0.01 seconds". The footer of the interface shows "Copyright © 1999, 2022, Oracle and/or its affiliates." and "Oracle APEX 22.21".

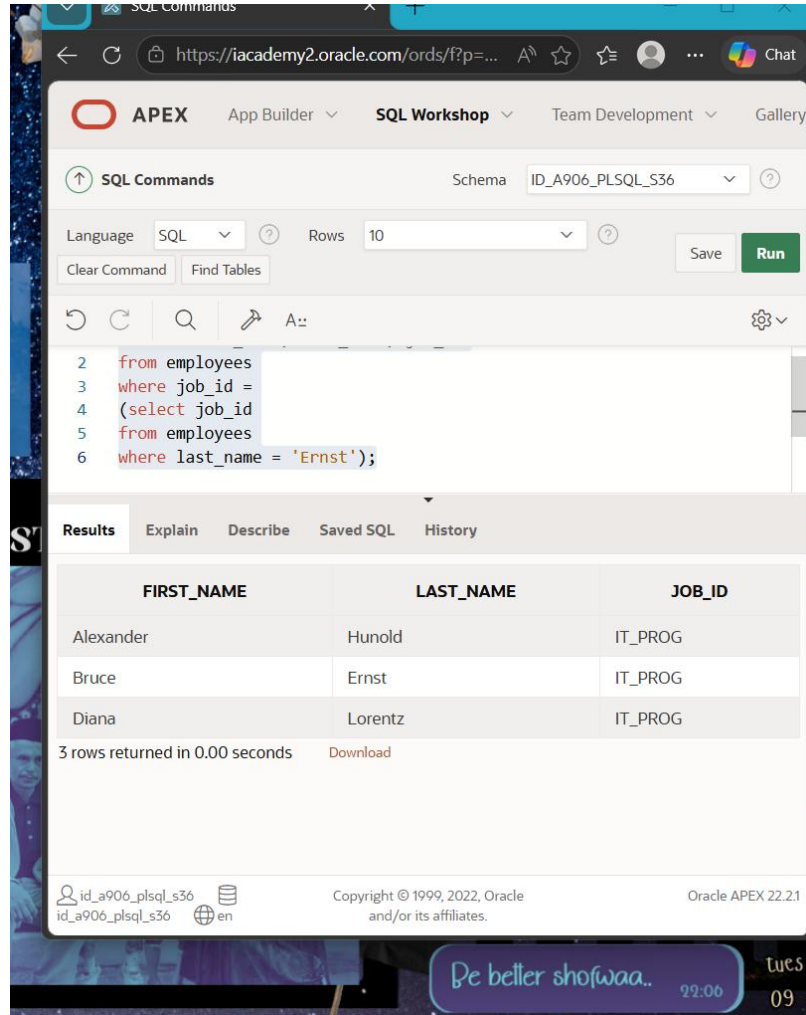
i. Halaman 22

i. Query:

```
select first_name, last_name, job_id
from employees
where job_id =
(select job_id
from employees
where last_name = 'Ernst');
```

- ii. Penjelasan:
Menampilkan first name, last name dan job id dari tabel employee, di mana job id berada di last name = 'Ernst'

- iii. Output:



The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command entered is:

```
2 from employees
3 where job_id =
4 (select job_id
5 from employees
6 where last_name = 'Ernst');
```

The results are displayed in a table with the following data:

FIRST_NAME	LAST_NAME	JOB_ID
Alexander	Hunold	IT_PROG
Bruce	Ernst	IT_PROG
Diana	Lorentz	IT_PROG

3 rows returned in 0.00 seconds

- j. Halaman 24

- i. Query:
- ```
select first_name, last_name, job_id
from employees
where job_id in
(select job_id from employees
where last_name = 'Ernst');
```

- ii. Penjelasan:

Menampilkan first name, last name dan job id dari tabel employee dengan job id dari employee di mana last name = 'Ernst'

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL Commands tab is active, displaying the following query:

```

1 select first_name, last_name, job_id
2 from employees
3 where job_id in
4 (select job_id from employees
5 where last_name = 'Ernst');

```

The Results tab is selected, showing the output of the query. The results are displayed in a table with three columns: FIRST\_NAME, LAST\_NAME, and JOB\_ID. The table contains three rows of data:

| FIRST_NAME | LAST_NAME | JOB_ID  |
|------------|-----------|---------|
| Alexander  | Hunold    | IT_PROG |
| Bruce      | Ernst     | IT_PROG |
| Diana      | Lorentz   | IT_PROG |

Below the table, it indicates "3 rows returned in 0.01 seconds" and provides a "Download" link. The footer of the interface shows the user "id\_a906\_plsql\_s36", the copyright "Copyright © 1999, 2022, Oracle and/or its affiliates.", and the version "Oracle APEX 22.21".

4. Section 10.4

a. Halaman 9

i. Query:

```

select o.first_name, o.last_name, o.salary
from employees o
where o.salary >
(select AVG (i.salary)
from employees i

```

where i.department\_id =  
o.department\_id);

ii. Penjelasan:

Menampilkan first name, last name dan salary dari tabel employee dimana jumlah salary berasal dari i.department\_id = o.department\_id

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL Commands tab is active, displaying a query that filters employees by department and calculates the average salary. The query is as follows:

```
3 where o.salary >
4 (select AVG (i.salary)
5 from employees i
6 where i.department_id =
7 o.department_id);
```

The Results tab shows the output of the query, which is a table with three columns: FIRST\_NAME, LAST\_NAME, and SALARY. The table contains six rows of data:

| FIRST_NAME | LAST_NAME | SALARY |
|------------|-----------|--------|
| Michael    | Hartstein | 13000  |
| Kevin      | Mourgos   | 5800   |
| Alexander  | Hunold    | 9000   |
| Eleni      | Zlotkey   | 10500  |
| Ellen      | Abel      | 11000  |
| Steven     | King      | 24000  |

The interface also shows the schema ID\_A906\_PLSQL\_S36, the language set to SQL, and the number of rows set to 10. The footer indicates the copyright is © 1999, 2022, Oracle and/or its affiliates, and the version is Oracle APEX 22.2.1.

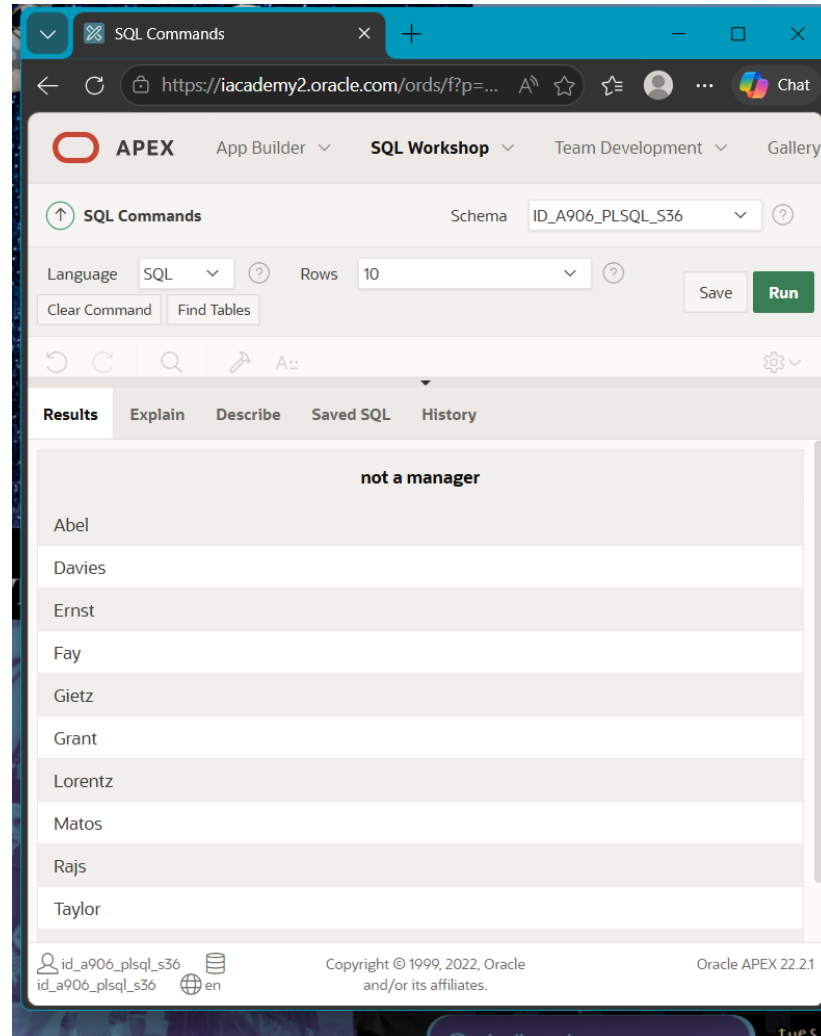
b. Halaman 13

i. Query:

```
select last_name as "not a manager"
from employees emp
where not exists
(select *
from employees mgr
where mgr.manager_id = emp.employee_id);
```

ii. Penjelasan:  
Menampilkan last name dengan nama tabel “not a manager” dari tabel employee.

iii. Output:



c. Halaman 14

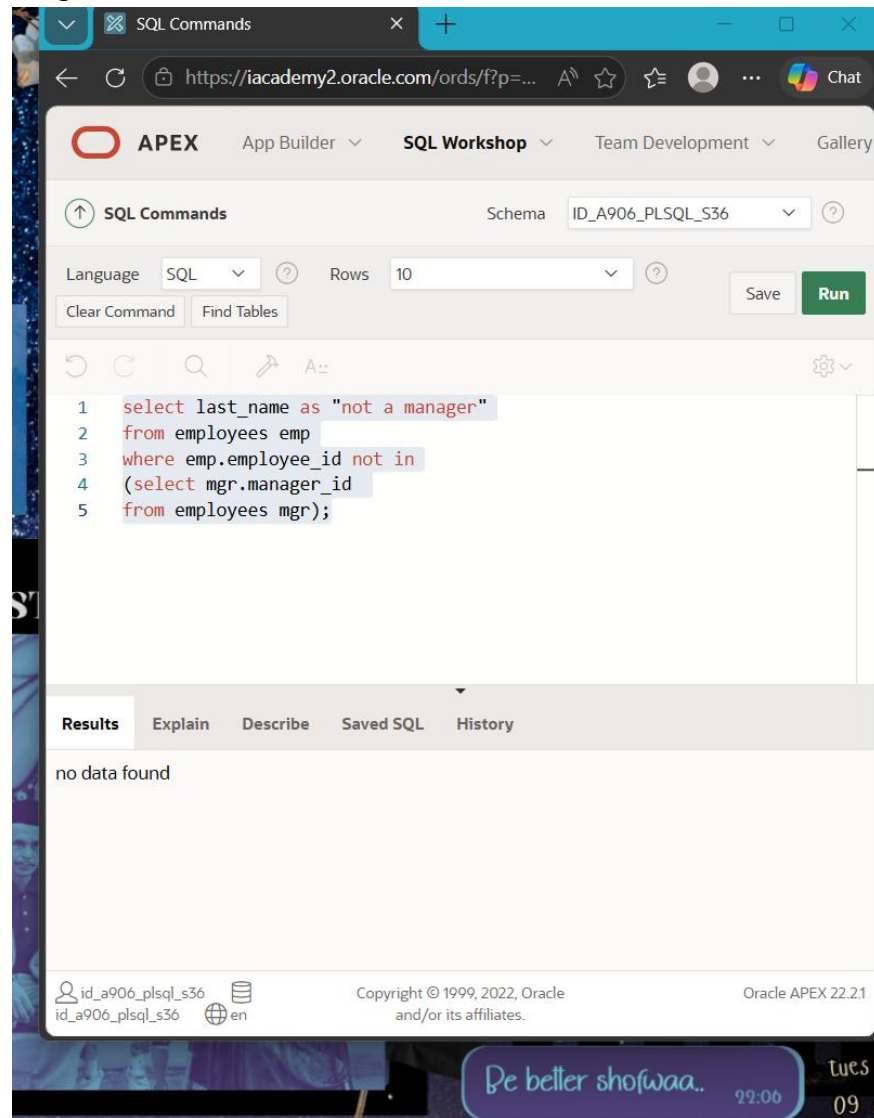
i. Query:  

```
select last_name as "not a manager"
from employees emp
where emp.employee_id not in
(select mgr.manager_id
from employees mgr);
```

ii. Penjelasan:

Menampilkan last name dengan tabel bernama not a manager dari tabel employee. Di mana manager id not in dari manager ID dari tabel employee

iii. Output:



d. Halaman 20

i. Query:

with managers as  
(select distinct manager\_id  
from employees  
where manager\_id is not null)

select last\_name as "Not a manager"

```
from employees
where employee_id not in
(select * from managers);
```

ii. Penjelasan:

Menampilkan last name dengan tabel yang bernama not a manager dari tabel employee. Di mana employee id not in select\* from managers.

iii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The query is executed, and the results are displayed in a table. The table has a title 'Not a manager' and contains 10 rows of data. The columns are not explicitly named, but the data represents the last names of employees who are not managers. The interface includes a 'SQL Commands' tab, a 'Schema' dropdown set to 'ID\_A906\_PLSQL\_S36', and a 'Rows' selector set to 10. The 'Run' button is highlighted in green. The bottom of the screen shows the user 'id\_a906\_plsql\_s36' and the Oracle APEX version '22.2.1'.

| Not a manager |
|---------------|
| Abel          |
| Davies        |
| Ernst         |
| Fay           |
| Gietz         |
| Grant         |
| Lorentz       |
| Matos         |
| Rajs          |
| Taylor        |

More than 10 rows available. Increase rows selector to view more rows.

5. Section 11.1

a. Halaman 5

i. Query:

```
select table_name
from user_tables
where table_name like 'JO%';
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, showing a query editor with the following SQL code:

```
1 select table_name
2 from user_tables
3 where table_name like 'JO%';
```

The query is set to run on the 'ID\_A906\_PLSQL\_S36' schema. The 'Language' is set to 'SQL' and 'Rows' is set to '10'. The 'Run' button is highlighted in green. Below the query editor, the 'Results' tab is selected, displaying the following table:

| TABLE_NAME  |
|-------------|
| JOBS        |
| JOB_GRADES  |
| JOB_HISTORY |

Below the table, it states '3 rows returned in 0.00 seconds' and provides a 'Download' link. The footer of the interface shows the user 'id\_a906\_plsql\_s36', the copyright 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'.

b. Halaman 6

i. Query:

```
select substr (first_name,1,1) || ' '|| last_name AS employee_name
from employees;
```

ii. Output:

The screenshot displays the Oracle APEX SQL Workshop interface. At the top, the browser address bar shows the URL `https://iacademy2.oracle.com/ords/f?p=...`. The APEX header includes the 'APEX' logo and navigation links for 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, showing a schema dropdown set to 'ID\_A906\_PLSQL\_S36'. The language is set to 'SQL' and the number of rows to display is '10'. Below the command area, there are buttons for 'Clear Command', 'Find Tables', 'Save', and 'Run'. The 'Run' button has been clicked, and the 'Results' tab is now selected. The results are displayed in a table with the column header 'EMPLOYEE\_NAME'. The table contains 10 rows of employee names, starting with 'E Abel' and ending with 'A Hunold'. At the bottom of the interface, there is a footer with user information 'id\_a906\_plsql\_s36', a globe icon, copyright text 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'. A banner at the very bottom reads 'Be better shofwaa..' with a timer '22:06' and the date 'TUES 09'.

| EMPLOYEE_NAME |
|---------------|
| E Abel        |
| C Davies      |
| L De Haan     |
| B Ernst       |
| P Fay         |
| W Gietz       |
| K Grant       |
| M Hartstein   |
| S Higgins     |
| A Hunold      |

c. Halaman 7

i. Query:

```
select first_name || ' ' || last_name AS employee_name, email
from employees
where email like '%IN%';
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, showing a query editor with the following SQL code:

```
1 select first_name || ' ' || last_name AS employee_name, email
2 from employees
3 where email like '%IN%';
```

The 'Schema' dropdown is set to 'ID\_A906\_PLSQL\_S36'. The 'Language' is 'SQL' and 'Rows' is set to 10. The 'Run' button is highlighted. Below the query editor, the 'Results' tab is active, displaying a table with two columns: 'EMPLOYEE\_NAME' and 'EMAIL'.

| EMPLOYEE_NAME  | EMAIL    |
|----------------|----------|
| ShelleyHiggins | SHIGGINS |
| StevenKing     | SKING    |

Below the table, it states '2 rows returned in 0.01 seconds' and provides a 'Download' link. The footer includes the user 'id\_a906\_plsql\_s36', the copyright notice 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'. A banner at the bottom right says 'Be better shofwaa.. 22:00'.

d. Halaman 8

i. Query:

```
select min (last_name) as "Smallest Last name",
max (last_name) as "Highest last name"
from employees;
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, showing a query in the 'SQL Commands' section. The schema is set to 'ID\_A906\_PLSQL\_S36'. The language is 'SQL' and the number of rows is set to '10'. The query is:

```
1 select min (last_name) as "Smallest Last name",
2 max (last_name) as "Highest last name"
3 from employees;
```

The 'Results' tab is selected, displaying a table with two columns: 'Smallest Last name' and 'Highest last name'. The results show 'Abel' for the smallest last name and 'Zlotkey' for the highest last name. Below the table, it states '1 rows returned in 0.01 seconds' and provides a 'Download' link. The footer includes the user 'id\_a906\_plsql\_s36', the copyright 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'.

| Smallest Last name | Highest last name |
|--------------------|-------------------|
| Abel               | Zlotkey           |

1 rows returned in 0.01 seconds [Download](#)

e. Halaman 9

i. Query:

```
select to_char (salary/4, '$9999.99') as weekly_salary
from employees
where salary/4 between 700 and 3000;
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' tab is active, showing a schema dropdown set to 'ID\_A906\_PLSQL\_S36'. The language is set to 'SQL' and the number of rows to display is '10'. The 'Run' button is highlighted in green. Below the command area, the 'Results' tab is selected, displaying a table with the title 'WEEKLY\_SALARY'. The table contains 10 rows of salary values formatted as currency. At the bottom, the footer shows the user 'id\_a906\_plsql\_s36', the copyright 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'. A banner at the very bottom reads 'Be better shofwaa.. 22:06'.

| WEEKLY_SALARY |
|---------------|
| \$1100.00     |
| \$3000.00     |
| \$2075.00     |
| \$2625.00     |
| \$2750.00     |
| \$2150.00     |
| \$1750.00     |
| \$1450.00     |
| \$875.00      |
| \$775.00      |

f. Halaman 10

i. Query:

```
select e.first_name,
 e.last_name,
 j.job_title
from employees e
join jobs j
on e.job_id = j.job_id
order by j.job_title;
```

ii. Output:

SQL Commands

<https://iacademy2.oracle.com/ords/f?p=...>

APEX

App Builder

SQL Workshop

Team Development

Gallery

SQL Commands

Schema

ID\_A906\_PLSQL\_S36

Language

SQL

Rows

10

Save

Run

Clear Command

Find Tables

Results

Explain

Describe

Saved SQL

History

| FIRST_NAME | LAST_NAME | JOB_TITLE                     |
|------------|-----------|-------------------------------|
| Shelley    | Higgins   | Accounting Manager            |
| Jennifer   | Whalen    | Administration Assistant      |
| Lex        | De Haan   | Administration Vice President |
| Neena      | Kochhar   | Administration Vice President |
| Michael    | Hartstein | Marketing Manager             |
| Pat        | Fay       | Marketing Representative      |
| Steven     | King      | President                     |
| Alexander  | Hunold    | Programmer                    |
| Bruce      | Ernst     | Programmer                    |
| Diana      | Lorentz   | Programmer                    |

More than 10 rows available. Increase rows selector to view more rows.

id\_a906\_plsql\_s36

id\_a906\_plsql\_s36

en

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Oracle APEX 22.2.1

Be better shofwaa..

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g. Halaman 11

i. Query:

```
select e.first_name,
 e.last_name,
 j.job_title,
 j.min_salary || ' - ' || j.max_salary AS salary_range,
 e.salary
from employees e
join jobs j
on e.job_id = j.job_id;
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' section shows the schema 'ID\_A906\_PLSQL\_S36' and the language 'SQL'. The 'Rows' selector is set to 10. The 'Run' button is highlighted. Below the command editor, the 'Results' tab is active, displaying a table with 10 rows of data. The table columns are: first\_name, last\_name, job\_title, salary\_range, and salary. The data includes employees like William Gietz, Shelley Higgins, Jennifer Whalen, Steven King, Lex De Haan, Neena Kochhar, Alexander Hunold, Diana Lorentz, Bruce Ernst, and Michael Hartstein. A message at the bottom of the table indicates 'More than 10 rows available. Increase rows selector to view more rows.' The footer shows '10 rows returned in 0.01 seconds', a 'Download' button, and copyright information for Oracle APEX 22.2.1.

| first_name | last_name | job_title                     | salary_range  | salary |
|------------|-----------|-------------------------------|---------------|--------|
| William    | Gietz     | Public Accountant             | 4200 - 9000   | 8300   |
| Shelley    | Higgins   | Accounting Manager            | 8200 - 16000  | 12000  |
| Jennifer   | Whalen    | Administration Assistant      | 3000 - 6000   | 4400   |
| Steven     | King      | President                     | 20000 - 40000 | 24000  |
| Lex        | De Haan   | Administration Vice President | 15000 - 30000 | 17000  |
| Neena      | Kochhar   | Administration Vice President | 15000 - 30000 | 17000  |
| Alexander  | Hunold    | Programmer                    | 4000 - 10000  | 9000   |
| Diana      | Lorentz   | Programmer                    | 4000 - 10000  | 4200   |
| Bruce      | Ernst     | Programmer                    | 4000 - 10000  | 6000   |
| Michael    | Hartstein | Marketing Manager             | 9000 - 15000  | 13000  |

h. Halaman 12

i. Query:

```
select substr (e.first_name,1,1) || " || e.last_name as employee_name,
d.department_name
from employees e
join departments d
on e.department_id = d.department_id;
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The SQL command entered is: `select substr (e.first_name,1,1) || " || e.last_name as employee_name, d.department_name from employees e join departments d on e.department_id = d.department_id;`. The results are displayed in a table with two columns: **EMPLOYEE\_NAME** and **DEPARTMENT\_NAME**. The table contains 10 rows of data. Below the table, it states "More than 10 rows available. Increase rows selector to view more rows." The interface also shows the schema as `ID_A906_PLSQL_S36`, the language as `SQL`, and the number of rows as `10`. The footer includes the Oracle APEX 22.2.1 version and a copyright notice.

| EMPLOYEE_NAME | DEPARTMENT_NAME |
|---------------|-----------------|
| JWhalen       | Administration  |
| MHartstein    | Marketing       |
| PFay          | Marketing       |
| CDavies       | Shipping        |
| PVargas       | Shipping        |
| TRajs         | Shipping        |
| KMourgos      | Shipping        |
| RMatos        | Shipping        |
| AHunold       | IT              |
| BErnst        | IT              |

i. Halaman 13

ii. Query:

```
select substr (e.first_name,1,1) || " || e.last_name as employee_name,
d.department_name
from employees e
join departments d
using (department_id);
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' section is active, showing a schema of 'ID\_A906\_PLSQL\_S36'. The language is set to 'SQL' and the number of rows to display is '10'. The query being executed is:

```
1 select substr (e.first_name,1,1) || ' ' || e.last_name as employee_name,
2 d.department_name
3 from employees e
4 join departments d
5 using (department_id);
```

The 'Results' tab is selected, displaying a table with two columns: 'EMPLOYEE\_NAME' and 'DEPARTMENT\_NAME'. The table contains the following data:

| EMPLOYEE_NAME | DEPARTMENT_NAME |
|---------------|-----------------|
| JWhalen       | Administration  |
| MHartstein    | Marketing       |
| PFay          | Marketing       |
| CDavies       | Shipping        |
| PVargas       | Shipping        |
| TRajs         | Shipping        |
| KMourgos      | Shipping        |
| RMatos        | Shipping        |
| AHunold       | IT              |

The bottom of the interface shows the user 'id\_a906\_plsql\_s36', the environment 'en', and the Oracle APEX 22.2.1 version. A footer banner at the bottom reads 'Be better shofwaa.. 22:06'.

j. Halaman 14

i. Query:

```
select decode (manager_id, null, 'nobody', 'somebody') as
manager_status, last_name
from employees;
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' section is active, showing the schema 'ID\_A906\_PLSQL\_S36'. The language is set to 'SQL' and the number of rows to display is '10'. The 'Run' button is highlighted in green. Below the command area, there are tabs for 'Results', 'Explain', 'Describe', 'Saved SQL', and 'History'. The 'Results' tab is selected, displaying a table with two columns: 'MANAGER\_STATUS' and 'LAST\_NAME'. The table contains 10 rows of data. At the bottom of the results area, a message states 'More than 10 rows available. Increase rows selector to view more rows.' and '10 rows returned in 0.01 seconds'. The footer of the interface shows the user 'id\_a906\_plsql\_s36', the URL 'iacademy2.oracle.com/ords/f...', the copyright notice 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'.

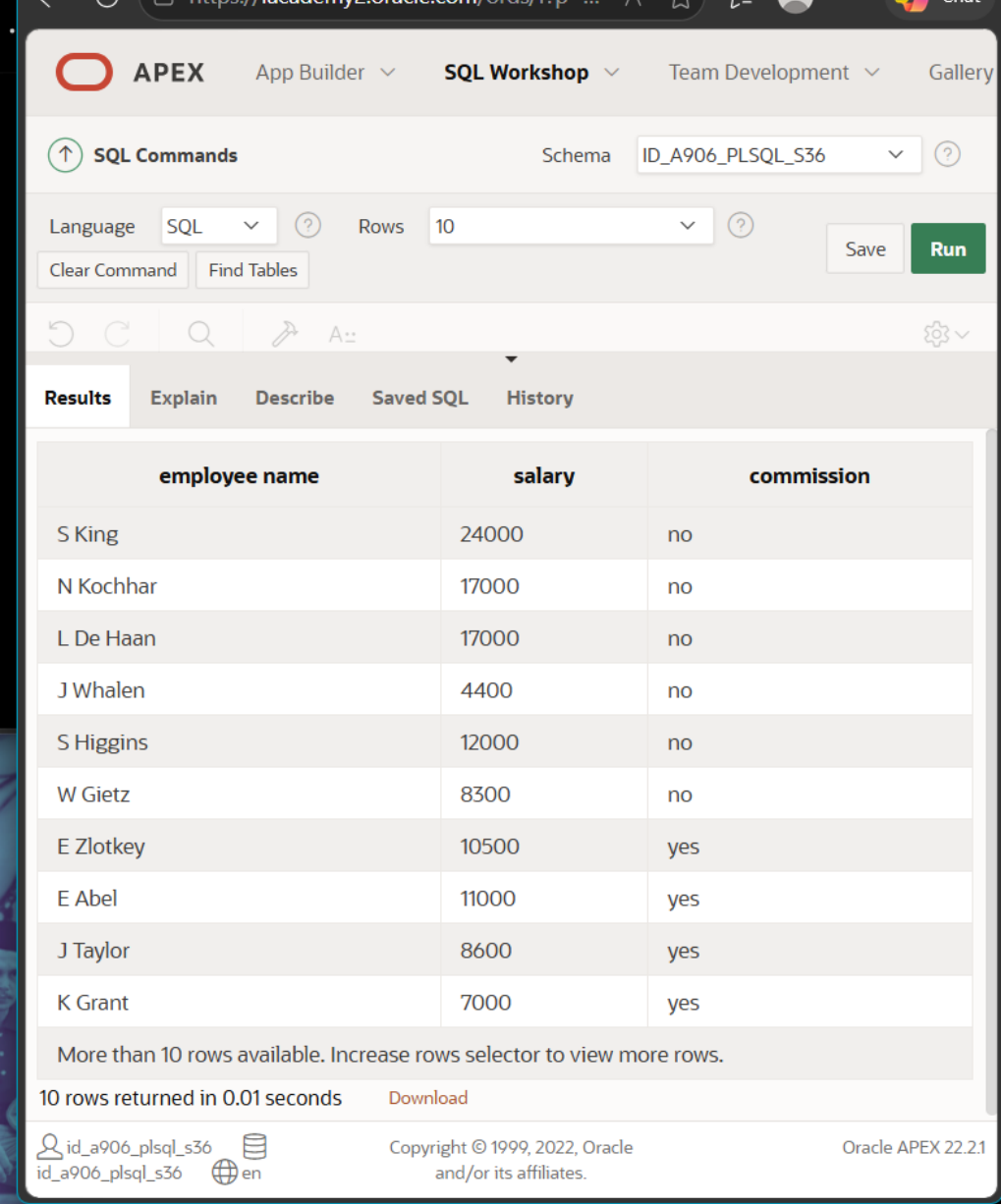
| MANAGER_STATUS | LAST_NAME |
|----------------|-----------|
| nobody         | King      |
| somebody       | Kochhar   |
| somebody       | De Haan   |
| somebody       | Whalen    |
| somebody       | Higgins   |
| somebody       | Gietz     |
| somebody       | Zlotkey   |
| somebody       | Abel      |
| somebody       | Taylor    |
| somebody       | Grant     |

k. Halaman 15

i. Query:

```
select substr (first_name,1,1) || ' ' || last_name as "employee name",
salary as "salary",
decode (commission_pct,
null, 'no','yes')as "commission"
from employees;
```

ii. Output:



The screenshot shows the Oracle APEX SQL Workshop interface. The query is executed against the schema ID\_A906\_PLSQL\_S36. The results are displayed in a table with 10 rows. The table has three columns: employee name, salary, and commission. The results show the first 10 rows of the employees table.

| employee name | salary | commission |
|---------------|--------|------------|
| S King        | 24000  | no         |
| N Kochhar     | 17000  | no         |
| L De Haan     | 17000  | no         |
| J Whalen      | 4400   | no         |
| S Higgins     | 12000  | no         |
| W Gietz       | 8300   | no         |
| E Zlotkey     | 10500  | yes        |
| E Abel        | 11000  | yes        |
| J Taylor      | 8600   | yes        |
| K Grant       | 7000   | yes        |

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

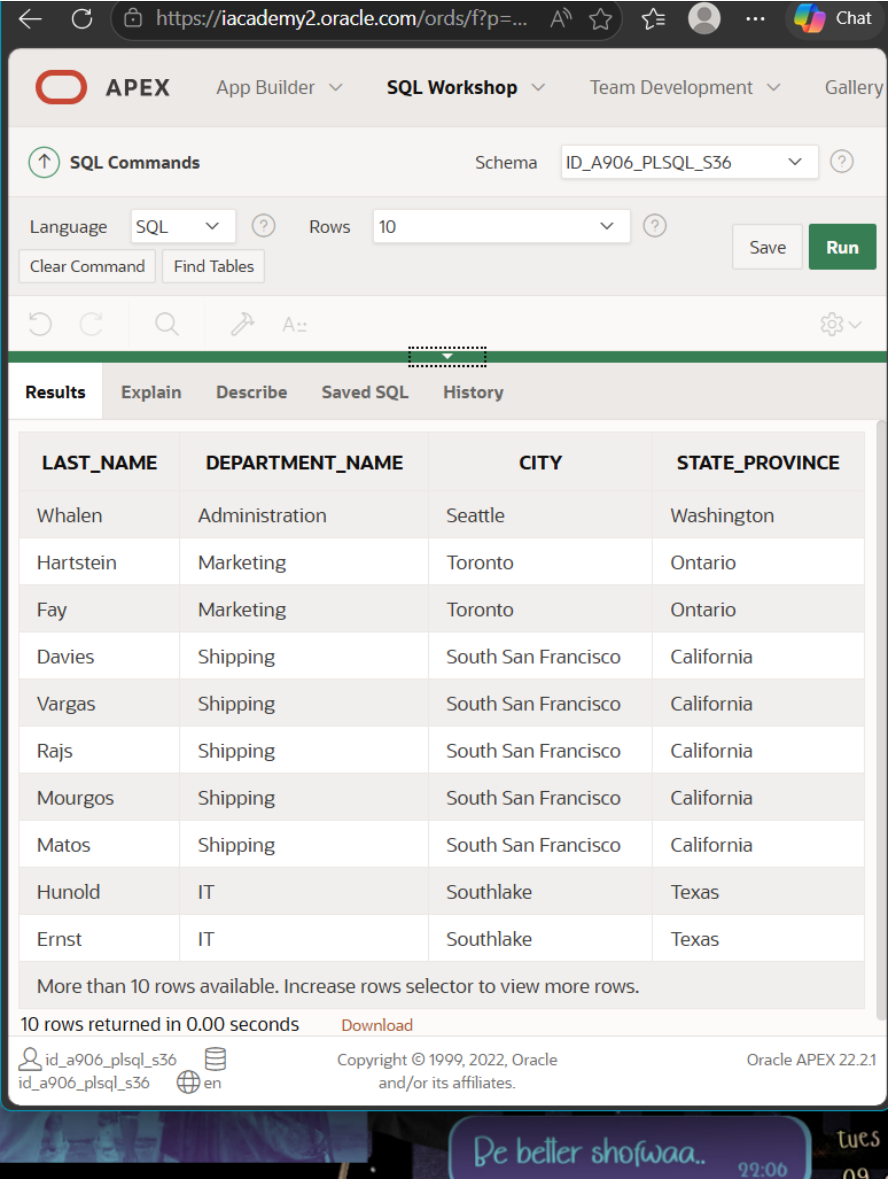
id\_a906\_plsql\_s36 en Copyright © 1999, 2022, Oracle and/or its affiliates. Oracle APEX 22.2.1

1. Halaman 16

i. Query:

```
select e.last_name,d.department_name,
l.city,
l.state_province
from departments d
left outer join employees e
on d.department_id = e.department_id
join locations l
on d.location_id = l.location_id;
```

ii. Output:



The screenshot displays the Oracle APEX SQL Workshop interface. At the top, the 'SQL Commands' section shows the schema 'ID\_A906\_PLSQL\_S36'. The language is set to 'SQL' and the number of rows to display is '10'. Below this, the 'Results' tab is active, showing a table with the following data:

| LAST_NAME | DEPARTMENT_NAME | CITY                | STATE_PROVINCE |
|-----------|-----------------|---------------------|----------------|
| Whalen    | Administration  | Seattle             | Washington     |
| Hartstein | Marketing       | Toronto             | Ontario        |
| Fay       | Marketing       | Toronto             | Ontario        |
| Davies    | Shipping        | South San Francisco | California     |
| Vargas    | Shipping        | South San Francisco | California     |
| Rajs      | Shipping        | South San Francisco | California     |
| Mourgos   | Shipping        | South San Francisco | California     |
| Matos     | Shipping        | South San Francisco | California     |
| Hunold    | IT              | Southlake           | Texas          |
| Ernst     | IT              | Southlake           | Texas          |

Below the table, it states 'More than 10 rows available. Increase rows selector to view more rows.' and '10 rows returned in 0.00 seconds'. The footer shows the user 'id\_a906\_plsql\_s36', the version 'Oracle APEX 22.2.1', and the copyright 'Copyright © 1999, 2022, Oracle and/or its affiliates.'

m. Halaman 17

i. Query:

```
select first_name, last_name,
 nvl (to_char (commission_pct),
 nvl (to_char (manager_id), '-1')) as result
from employees;
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' section is active, showing the schema 'ID\_A906\_PLSQL\_S36'. The language is set to 'SQL' and the number of rows to display is '10'. The 'Run' button is highlighted. Below the command area, the 'Results' tab is selected, displaying a table with three columns: 'FIRST\_NAME', 'LAST\_NAME', and 'RESULT'. The table contains 10 rows of data. At the bottom, it indicates '10 rows returned in 0.01 seconds' and provides a 'Download' link. The footer shows the user 'id\_a906\_plsql\_s36', the copyright 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'.

| FIRST_NAME | LAST_NAME | RESULT |
|------------|-----------|--------|
| Steven     | King      | -1     |
| Neena      | Kochhar   | 100    |
| Lex        | De Haan   | 100    |
| Jennifer   | Whalen    | 101    |
| Shelley    | Higgins   | 101    |
| William    | Gietz     | 205    |
| Eleni      | Zlotkey   | .2     |
| Ellen      | Abel      | .3     |
| Jonathon   | Taylor    | .2     |
| Kimberely  | Grant     | .15    |

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

id\_a906\_plsql\_s36  
id\_a906\_plsql\_s36 en

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Be better shofwaa.. 22:06

tues 09 d

n. Halaman 18

i. Query:

```
select e.last_name, e.salary, jg.grade_level as job_grade
from employees e
join job_grades jg
on e.salary between jg.lowest_sal and jg.highest_sal
where e.department_id > 50;
```

ii. Output:

The screenshot displays the Oracle APEX SQL Workshop interface. At the top, the navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. Below this, the 'SQL Commands' section shows the schema 'ID\_A906\_PLSQL\_S36'. The 'Language' is set to 'SQL' and 'Rows' is set to '10'. The 'Run' button is highlighted in green. The SQL command entered is: `1 select e.last_name, e.salary, jg.grade_level as job_grade`. Below the command, the 'Results' tab is active, displaying a table with 10 rows. The table columns are last\_name, salary, and job\_grade. The results show employees with department\_id > 50, including Lorentz, Ernst, Gietz, Taylor, Hunold, Zlotkey, Abel, Higgins, Kochhar, and De Haan. At the bottom, it indicates '10 rows returned in 0.01 seconds' and provides a 'Download' link. The footer shows the user 'id\_a906\_plsql\_s36', the copyright 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'.

| last_name | salary | job_grade |
|-----------|--------|-----------|
| Lorentz   | 4200   | B         |
| Ernst     | 6000   | C         |
| Gietz     | 8300   | C         |
| Taylor    | 8600   | C         |
| Hunold    | 9000   | C         |
| Zlotkey   | 10500  | D         |
| Abel      | 11000  | D         |
| Higgins   | 12000  | D         |
| Kochhar   | 17000  | E         |
| De Haan   | 17000  | E         |

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

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o. Halaman19

i. Query:

```
select e.last_name, d.department_name
from employees e
full outer join departments d
on e.department_id = d.department_id;
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. At the top, there's a navigation bar with 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. Below this, the 'SQL Commands' section is active, showing a schema dropdown set to 'ID\_A906\_PLSQL\_S36'. The language is set to 'SQL' and the number of rows to display is '10'. The query being executed is:

```
1 select e.last_name, d.department_name
2 from employees e
3 full outer join departments d
```

Below the query, there are tabs for 'Results', 'Explain', 'Describe', 'Saved SQL', and 'History'. The 'Results' tab is selected, displaying a table with two columns: 'LAST\_NAME' and 'DEPARTMENT\_NAME'. The table contains 11 rows of data:

| LAST_NAME | DEPARTMENT_NAME |
|-----------|-----------------|
| King      | Executive       |
| Kochhar   | Executive       |
| De Haan   | Executive       |
| Whalen    | Administration  |
| Higgins   | Accounting      |
| Gietz     | Accounting      |
| Zlotkey   | Sales           |
| Abel      | Sales           |
| Taylor    | Sales           |
| Grant     | -               |

At the bottom of the interface, there's a footer with user information 'id\_a906\_plsql\_s36', a globe icon, copyright notice 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and 'Oracle APEX 22.2.1'. A blue banner at the very bottom says 'Be better shofwaa..' with a timer '22:00' and 'Tues 09'.

p. Halaman 20

i. Query:

```
select e.last_name as employee,
prior e.last_name as manager,
level as position
from employees e
start with employee_id = 100
connect by prior employee_id = manager_id;
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The 'SQL Commands' section displays the query: `select e.last_name as employee, prior e.last_name as manager, level as position from employees e start with employee_id = 100 connect by prior employee_id = manager_id;`. The 'Results' tab is active, showing a table with 10 rows. The table has three columns: 'EMPLOYEE', 'MANAGER', and 'POSITION'. The data shows a hierarchy starting from King (position 1) and branching out to other employees like Kochhar, Whalen, Higgins, Gietz, De Haan, Hunold, Ernst, Lorentz, and Mourgos. The bottom of the screen shows the footer with '10 rows returned in 0.01 seconds', 'Download' link, and 'Oracle APEX 22.2.1'.

| EMPLOYEE | MANAGER | POSITION |
|----------|---------|----------|
| King     | -       | 1        |
| Kochhar  | King    | 2        |
| Whalen   | Kochhar | 3        |
| Higgins  | Kochhar | 3        |
| Gietz    | Higgins | 4        |
| De Haan  | King    | 2        |
| Hunold   | De Haan | 3        |
| Ernst    | Hunold  | 4        |
| Lorentz  | Hunold  | 4        |
| Mourgos  | King    | 2        |

More than 10 rows available. Increase rows selector to view more rows.

10 rows returned in 0.01 seconds [Download](#)

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Be better shofwaa.. 22:06 Tues 09

q. Halaman21

i. Query:

```
select min (hire_date)as earliest_hire_dfate,
max(hire_date) as latest_hire_date,
count (*) as number_of_employees
from employees;
```

ii. Output:

The screenshot shows the Oracle APEX SQL Workshop interface. The top navigation bar includes 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. The 'SQL Commands' section is active, showing a schema of 'ID\_A906\_PLSQL\_S36'. The language is set to 'SQL' and the number of rows to display is '10'. The query being executed is:

```
1 select min (hire_date)as earliest_hire_dfate,
2 max(hire_date) as latest_hire_date,
3 count (*) as number_of_employees
4 from employees;
```

The 'Results' tab is selected, displaying a table with the following data:

| EARLIEST_HIRE_DFATE | LATEST_HIRE_DATE | NUMBER_OF_EMPLOYEES |
|---------------------|------------------|---------------------|
| 17-Jun-1987         | 29-Jan-2000      | 20                  |

Below the table, it states '1 rows returned in 0.01 seconds' and provides a 'Download' link. The footer of the interface includes the user 'id\_a906\_plsql\_s36', the language 'en', the copyright notice 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'. A banner at the bottom right says 'Be better shofwaa.. 22:06'.

r. Halaman 22

i. Query:

```
select d.department_name, sum(e.salary) as departmental_cost
from employees e
join departments d
on e.department_id = d.department_id
group by d.department_name
having sum (e.salary) between 15000 and 31000
order by departmental_cost;
```

ii. Output:

The screenshot shows the APEX SQL Workshop interface. At the top, there's a navigation bar with 'APEX', 'App Builder', 'SQL Workshop', 'Team Development', and 'Gallery'. Below this, the 'SQL Commands' section is active, showing a schema dropdown set to 'ID\_A906\_PLSQL\_S36'. The language is set to 'SQL' and the number of rows to display is '10'. The query entered is: `select d.department_name, sum(e.salary) as departmental_cost`. Below the query, there are tabs for 'Results', 'Explain', 'Describe', 'Saved SQL', and 'History'. The 'Results' tab is selected, displaying a table with two columns: 'DEPARTMENT\_NAME' and 'DEPARTMENTAL\_COST'. The table contains five rows of data. At the bottom of the results section, it states '5 rows returned in 0.00 seconds' and provides a 'Download' link. The footer of the interface includes the user 'id\_a906\_plsql\_s36', the copyright notice 'Copyright © 1999, 2022, Oracle and/or its affiliates.', and the version 'Oracle APEX 22.2.1'.

| DEPARTMENT_NAME | DEPARTMENTAL_COST |
|-----------------|-------------------|
| Shipping        | 17500             |
| Marketing       | 19000             |
| IT              | 19200             |
| Accounting      | 20300             |
| Sales           | 30100             |

s. Halaman 23

i. Query:

```
select d.department_name, d.manager_id, m.last_name AS
manager_name,
AVG(e.salary) AS average_salary
from departments d
join employees e
on d.department_id = e.department_id
join employees m
on d.manager_id = m.employee_id
group by d.department_name, d.manager_id, m.last_name;
```

ii. Output:

The screenshot displays the Oracle APEX SQL Workshop interface. The 'SQL Commands' tab is active, showing a query that selects department names, manager IDs, manager names, and the average salary for each department. The query is executed, and the 'Results' tab shows a table with 7 rows of data. The table columns are DEPARTMENT\_NAME, MANAGER\_ID, MANAGER\_NAME, and AVERAGE\_SALARY. The data rows are: Administration (200, Whalen, 4400), Sales (149, Zlotkey, 10033.3333333333333333333333333333), Marketing (201, Hartstein, 9500), Shipping (124, Mourgos, 3500), Accounting (205, Higgins, 10150), Executive (100, King, 19333.3333333333333333333333333333), and IT (103, Hunold, 6400). The interface also shows the number of rows returned (7 rows) and the execution time (0.02 seconds). The footer indicates the user is 'id\_a906\_plsql\_s36' and the version is 'Oracle APEX 22.21'.

| DEPARTMENT_NAME | MANAGER_ID | MANAGER_NAME | AVERAGE_SALARY                     |
|-----------------|------------|--------------|------------------------------------|
| Administration  | 200        | Whalen       | 4400                               |
| Sales           | 149        | Zlotkey      | 10033.3333333333333333333333333333 |
| Marketing       | 201        | Hartstein    | 9500                               |
| Shipping        | 124        | Mourgos      | 3500                               |
| Accounting      | 205        | Higgins      | 10150                              |
| Executive       | 100        | King         | 19333.3333333333333333333333333333 |
| IT              | 103        | Hunold       | 6400                               |

## BAB III

### KESIMPULAN

#### 3.1 Ringkasan Hasil Praktikum

Praktikum membahas penggunaan subquery pada database oracle menggunakan tabel employees dan departments. Materi meliputi penggunaan operator perbandingan, ANY, ALL, IN, EXIST, NOT EXIST serta correlated sub query untuk menampilkan data pegawai berdasarkan department, salary, manager, hire date dan job tertentu

#### 3.2 Saran Pengembangan

Praktikum selanjutnya dapat dikembangkan dengan penggunaan join, view, function, procedure dan optimasi query agar mahasiswa memahami pengelolaan database secara kompl

# DAFTAR PUSTAKA

1. Oracle Database SQL Language Reference. Oracle Corporation. *Oracle Database SQL Language Reference*. Oracle Documentation.
2. Database System Concepts. Silberschatz, A., Korth, H. F., & Sudarshan, S. *Database System Concepts*. McGraw-Hill.
3. Sistem Basis Data. Kadir, A. *Sistem Basis Data*. Penerbit Andi.
4. Fundamentals of Database Systems. Elmasri, R., & Navathe, S. B. *Fundamentals of Database Systems*. Pearson Education.
5. [Oracle Documentation](#)